Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Question

Full Marks: 20 *Deadline: 5:30 PM, 28-08-2025*

1. Write a Bash script to validate whether a given input is a valid IPv4 address.The script should: (a) Prompt the user to enter an IP address. (b) Use a regular expression to check that the input follows the correct IPv4 format: (i) Four groups of 1 to 3 digits (ii) Groups separated by dots (.)

If the format is valid, use string processing (e.g., cut) to:

- Split the IP into four octets
- Verify that each octet is a number between 0 and 255

Display:

- Valid IP address. if all checks pass
- Appropriate error messages if the format is invalid or any octet is out of range

Sample output:

```
arindam@ARINDAM:~$ ./ipv4_check.sh
Enter an IP address:
4444.22.3.1
Invalid IP format.
arindam@ARINDAM:~$ ./ipv4_check.sh
Enter an IP address:
242.455.10.2
Invalid IP: one or more octets are out of range (0-255).
arindam@ARINDAM:~$ ./ipv4_check.sh
Enter an IP address:
10.30.0.1
Valid IP address.
arindam@ARINDAM:~$
```

2. Write a Bash script that uses functions to find the maximum element in an array. Requirements:

- The script must define two functions:
- A main function that:
    - Prompts the user for the number of elements.
    - Reads the elements of the array from the user.
    - Prints the maximum element
- A greatest function that:
    - Takes the array as input (via arguments).
    - Calculates and returns the maximum element to the main

The main function should call the greatest function with the array as an argument